

MOSAIC: Multimodal Perceptual Cue Fusion for Full-Reference Point Cloud Quality Assessment

Jiaxin Li, Tong Chen, Zehong Li, Dandan Ding, and Zhan Ma

Abstract—Full-reference point cloud quality assessment (FR-PCQA) has access to the original point cloud, but reliable quality prediction still depends on how geometric, photometric, and view-dependent discrepancies are represented and fused. Motivated by David Marr’s three-level account of vision, we formulate FR-PCQA as a multimodal perceptual inference problem and propose MOSAIC, a training-free cue-fusion framework. MOSAIC extracts local 3D structural cues, 3D color cues, and projected 2D saliency cues from reference and distorted point clouds. Instead of aggregating all cues in a single stage, MOSAIC first combines geometry and color fidelity into an intrinsic 3D fidelity score, and then incorporates 2D saliency consistency to reflect attention-related perceptual importance. A bounded adaptive weighting strategy produces the final perceptual fidelity score without mean opinion score (MOS)-supervised training or regression-based score integration. Experiments on five public PCQA databases show that MOSAIC achieves the best average performance among training-free metrics with complete results, obtains the best results on several datasets and distortion groups, and remains competitive with representative learning-based models while avoiding supervised score fitting. Materials are accessible at <https://njuvision.github.io/MOSAIC>.

Index Terms—Point Cloud Quality Assessment, Multimodal Cue Fusion, 3D Geometry, 3D Color, 2D Saliency

I. INTRODUCTION

Point clouds have become a practical representation for immersive applications such as Virtual/Augmented/eXtended Reality (VR/AR/XR), where they compactly describe 3D objects and scenes. In real-world systems, however, acquisition, compression, transmission, and rendering can introduce visible geometry and color artifacts. Point cloud quality assessment (PCQA) therefore plays an important role in predicting subjective quality, such as the mean opinion score (MOS), and in guiding the optimization of end-to-end visual experience.

Depending on the availability of the original signal, PCQA methods are commonly categorized as full-reference (FR), reduced-reference (RR), or no-reference (NR). With complete reference information, FR-PCQA methods generally provide more stable and reliable quality prediction [1].

Complete reference information, however, does not specify how distortions should be represented or pooled. Direct geometry and color metrics are efficient, but they mainly capture point-wise, low-level local discrepancies and often treat geometry and appearance separately [2]–[4]. Subsequent metrics address this limitation by using local neighborhoods, graph or multiscale analysis, and saliency-aware local processing [1], [5]–[8]. Regression-based and deep-learning methods

can further improve correlation with subjective scores by learning score mappings [9]–[13], but their performance often depends on MOS-supervised fitting, specific training, or model design. Despite these advances, many methods still emphasize one primary type of perceptual evidence, such as 3D-domain discrepancies, projected 2D-view quality, or learned score mappings. This simplified cue modeling is not fully aligned with human visual perception.

Vision theory and psychophysical studies suggest that human vision recovers scene layout and surface structure by combining several cues, while attention affects which regions dominate quality judgments [14]–[17]. Point-cloud quality perception is therefore a mixed 3D–2D process: intrinsic 3D structure and attributes provide object-level fidelity, whereas 2D projections determine view-dependent visibility and attention. This raises the central question of this work: *how can complete reference information be organized into a training-free and interpretable fusion of 3D structural, 3D appearance, and 2D attention-related cues?*

To address this issue, we revisit FR-PCQA from Marr’s three-level perspective of vision [14]. At the computational level, the task is to infer perceived quality from the discrepancy between a reference and a distorted 3D representation. At the representational level, this discrepancy should be described by perceptually meaningful cues, including geometric structure, appearance consistency, and attention-related visual importance. At the implementational level, these cues can be realized through measurable descriptors extracted from point clouds and their projected views.

This motivates MOSAIC, a Multimodal Perceptual Cue Fusion framework that integrates intrinsic 3D cues with projected 2D saliency cues. Rather than developing new geometric descriptors, MOSAIC uses established descriptors as interpretable structural-fidelity evidence within the proposed multimodal cue organization. Density, color, and projected 2D saliency provide complementary evidence. MOSAIC then combines their modality-wise fidelity scores through a two-stage fusion process: it first integrates intrinsic 3D geometric structure and color appearance, and then incorporates attention-related saliency fidelity. Unlike saliency-assisted pooling methods, MOSAIC treats saliency consistency as a modality-level cue rather than only a spatial weighting factor.

The main contributions are summarized as follows:

- We formulate FR-PCQA from Marr’s three-level perspective as a multimodal perceptual cue-fusion problem. This formulation clarifies that perceived point cloud quality should be represented by complementary evidence from

TABLE I
FREQUENTLY USED NOTATIONS AND ACRONYMS IN THIS PAPER.

Term	Description
PCQA	Point cloud quality assessment
FR-PCQA	Full-reference point cloud quality assessment
HVS	Human visual system
MOS	Mean opinion score
MSE	Mean squared error
PSNR	Peak signal-to-noise ratio
D1 / D2	Point-to-point / point-to-plane geometric distance
KNN	K-nearest neighbors
SVR	Support vector regression
G-PCC / V-PCC	Geometry-based / video-based point cloud compression
RGB / YUV / LAB	Color spaces used for point-cloud attributes
PLCC / SROCC / RMSE	Correlation and error evaluation metrics

3D geometric structure, 3D color appearance, and 2D attention-related saliency, rather than by a single dominant cue in existing works.

- We propose MOSAIC, a training-free FR-PCQA metric that converts these perceptual cues into modality-wise fidelity scores. Instead of developing new geometric descriptors, MOSAIC uses established structural descriptors as interpretable geometric evidence and fuses them with color and projected 2D saliency through a two-stage adaptive rule.
- We conduct extensive evaluations on five public PCQA databases and distortion-wise benchmarks. MOSAIC achieves the best average performance among training-free metrics with complete results, obtains the best results on several datasets and distortion groups, and remains competitive with representative regression- and learning-based methods without MOS-supervised score fitting.

Frequently used notations in this work are listed in Table I.

II. RELATED WORK

Projection-based PCQA can reuse mature image quality assessment (IQA) tools by rendering point clouds into multiple 2D views [18]–[22]. However, the resulting scores are sensitive to viewpoint and rendering settings. Therefore, many FR-PCQA studies still investigate native 3D-space metrics that directly compare geometry and color discrepancies. These methods can be broadly grouped into training-free metrics and learning-based models.

A. Training-Free FR-PCQA

Training-free FR-PCQA metrics compute quality scores from reference-distorted differences without MOS-supervised fitting. Early metrics mainly measure direct point-wise discrepancies. Point-to-point (p2po) [2] and point-to-plane (p2pl) [3] distances quantify geometric errors, while luminance or color PSNR measures attribute distortion [4]. These metrics are simple and efficient, but geometry and color are treated largely as separate low-level signals.

Later methods moved from isolated point errors to local structural comparison. PCQM [5] extracts local geometric

and color descriptors from matched point neighborhoods and combines selected curvature- and lightness-related features into a handcrafted full-reference score. PointSSIM [23] extends structural similarity to point clouds by comparing local statistics of geometry, normal, curvature, or color features. GraphSIM [6] constructs local graphs around key points, treats color attributes as graph signals, and combines graph-based color similarity with geometric weighting; MS-GraphSIM [7] evaluates graph similarity at multiple neighborhood radii. MPED [24] describes local distortions through multiscale potential energy discrepancy. More recently, PQI [1] refined the same neighbor-to-neighbor paradigm by selecting scale-wise key points, constructing local neighborhoods with a fixed number of neighboring points, and measuring geometry and attribute similarity at multiple scales.

Local structural metrics describe how strongly geometry or color is degraded, while subjective quality also depends on whether the degraded regions are likely to attract visual attention. Motivated by this perception-related factor, PQSM [8] generates depth-enhanced 3D saliency maps from projected views, computes local saliency similarity together with local geometry and color similarities, multiplies the three similarities into a local quality index, and then applies saliency-based pooling. Thus, PQSM embeds saliency into a 3D neighborhood quality metric. MOSAIC organizes saliency differently: it keeps saliency in the projected-view domain, compares reference and distorted saliency maps as an independent modality-wise fidelity cue, and fuses this cue after intrinsic 3D geometry-color fidelity has been formed. This design separates native 3D degradation from view-dependent attention changes.

Beyond saliency-aware local-neighborhood metrics such as PQSM, recent methods further organize quality evidence through broader perceptual hypotheses. TCDM [25] formulates quality assessment through transformational complexity: it partitions the reference and distorted point clouds into local 3D Voronoi patch pairs and uses a space-aware vector autoregressive model to estimate self-complexity, transformational complexity, and prediction-related terms. PHM [26] adopts a perception-guided hybrid strategy according to distortion degree. It uses two branches: a visible-difference branch for high-quality samples based on texture-complexity-compensated luminance PSNR, and an appearance-degradation branch that combines graph signal variation with weighted co-occurrence statistics of spectral graph wavelet coefficients. Their contribution is not simply a stronger local descriptor, but a different organization of quality evidence: TCDM asks whether recovery complexity can reflect perceptual degradation, whereas PHM asks whether different distortion levels should trigger different HVS-inspired strategies.

B. Learning-Based FR-PCQA

Learning-based FR-PCQA methods use subjective labels or learned features to improve prediction. One line uses handcrafted descriptors as input features and learns a mapping to subjective scores. PointPCA [9] builds objective quality descriptors from PCA-based local neighborhood analysis

and uses them for quality regression. FRSVR [10] computes five full-reference metrics covering geometry distortion, color distortion, and point-count changes, and integrates them using support vector regression (SVR). SGW-PCQA+ [11] uses spectral graph wavelet representations with supervised score integration. More recently, the Feature Selection Model (FSM) [27] benchmarks features from representative FR metrics, applies recursive feature elimination with SVR or ridge regression, and combines selected features from PCQM, MS-GraphSIM, and the PSNR of D2-based MSE through ridge regression. These methods are useful references, but their scores depend on supervised fitting and evaluation splits.

Another line learns visual representations or quality mappings from point-cloud patches, projected views, point-cloud/mesh descriptors, or multiscale features [12], [28]–[30]. Tliba *et al.* [28] extract local deep features from reference and distorted point-cloud patches and measure visual degradation through cross-correlation between the learned embeddings. CT-PCQA [12] renders point clouds into projected views and extracts visual features with CNN and Transformer backbones, including ConvNeXt [31] and Swin Transformer [32]. The model supports both full-reference and no-reference prediction settings.

Large multimodal model (LMM)-based NR-PCQA has recently been explored in LMM-PCQA [33]. It converts quality labels into textual quality levels and combines LMM outputs with multiscale structural features. This idea inspired FR-MLLM, a full-reference method based on multimodal large language models (MLLMs) [13]. FR-MLLM investigates rendered-view prompting, response similarity, relative-quality response scores, fine-tuned absolute-quality scoring, and SVR-based score integration with conventional FR metrics. These LMM- and MLLM-based studies are promising because they bring high-level semantic and cross-modal priors into PCQA, but they also depend on rendering prompts, response parsing, model fine-tuning, and often supervised score integration.

C. Remarks

The above studies show that FR-PCQA has evolved from point-wise error measurement to local structural comparison, perceptual-hypothesis-driven modeling, and supervised representation learning. This progress also exposes a remaining gap. Training-free metrics are interpretable and reproducible, but they often organize quality evidence around one dominant axis, such as local geometry/color similarity, transformational complexity, or distortion-level-dependent graph analysis. Learning-based models can improve numerical correlation by fitting features to subjective labels, and recent models further introduce multimodal evidence through projected views or LMM/MLLM priors. However, these gains usually depend on training data or model-specific protocols, leaving multimodal cue organization less explicit in a training-free setting. MOSAIC addresses this setting by keeping the training-free and interpretable nature of handcrafted metrics while organizing multimodal perceptual evidence more explicitly. Specifically, MOSAIC first forms an intrinsic 3D fidelity estimate from geometry and color, and then incorporates projected 2D saliency consistency as a

view-dependent attention cue. This two-stage design makes the roles of 3D structure, 3D appearance, and 2D visual attention explicit, instead of merging all cues in a single-stage aggregation.

III. METHOD

MOSAIC follows Marr’s view by extracting perceptually meaningful cues of 3D geometric structure, 3D color appearance, and 2D visual saliency before fusing them for FR-PCQA. The overall framework is shown in Fig. 1. For a point cloud $\mathbf{P} = \{\vec{p}_1, \vec{p}_2, \dots, \vec{p}_N\} \in \mathbb{R}^{N \times 6}$, each point $\vec{p}_i$ contains 3D coordinates $\vec{g}_i$ and three-channel color attributes $\vec{c}_i$, i.e., $\vec{p}_i = \{\vec{g}_i, \vec{c}_i\} = \{(x_i, y_i, z_i), (r_i, g_i, b_i)\}$, $i \in [1, N]$. N is the total number of points.

A. Point Alignment

Given a reference point cloud $\mathbf{P}_r \in \mathbb{R}^{N \times 6}$ and a distorted point cloud $\mathbf{P}_d \in \mathbb{R}^{M \times 6}$, each point in $\mathbf{P}_r$ is treated as a key point. Its counterpart in $\mathbf{P}_d$ is identified by nearest-neighbor search using Euclidean distance in 3D space. The matched points form the aligned distorted point cloud $\mathbf{P}_{d'} \in \mathbb{R}^{N \times 6}$, which is used for subsequent point-wise feature comparison.

B. Multimodal Perceptual Representation

1) 3D Structural Representation via Geometry Features:

Local shape feature. Geometric distortions degrade local spatial structures such as surfaces and edges, thereby affecting perceptual quality. Following prior work [9], [34], [35], MOSAIC uses eigenvalue-based descriptors as interpretable 3D structural cues. These descriptors are not introduced as standalone novel shape features; rather, they provide a stable structural representation that can be compared between reference and distorted point clouds and then fused with color and saliency cues.

For each point $\vec{p}_i$, a local neighborhood $\{\vec{g}_i^k\}_{k=1}^{K_g}$ is constructed using K_g -nearest neighbors. The corresponding covariance matrix is computed as

$$\text{Cov}_i = \frac{1}{K_g} \sum_{k=1}^{K_g} (\vec{g}_i^k - \vec{g}_i) (\vec{g}_i^k - \vec{g}_i)^T. \quad (1)$$

Eigen-decomposition of Cov_i yields three ordered eigenvalues $\lambda_1 \geq \lambda_2 \geq \lambda_3$. Using these eigenvalues, the local curvature $G_i^{(1)}$ and linearity $G_i^{(2)}$ are defined as

$$G_i^{(1)} = \frac{\lambda_3}{\lambda_1 + \lambda_2 + \lambda_3}, \quad G_i^{(2)} = \frac{\lambda_1 - \lambda_2}{\lambda_1}. \quad (2)$$

The same descriptors are extracted from both the reference and resampled distorted point clouds, denoted as $\{G_{r_i}^{(1)}, G_{r_i}^{(2)}\}$ and $\{G_{d'i}^{(1)}, G_{d'i}^{(2)}\}$, respectively.

Local point density feature. Geometric resampling or noise changes both point positions and local density. To capture this effect, we compute the squared Euclidean distances between each point and its K_p nearest neighbors, and use their average as the local point density feature for both reference and distorted point clouds:

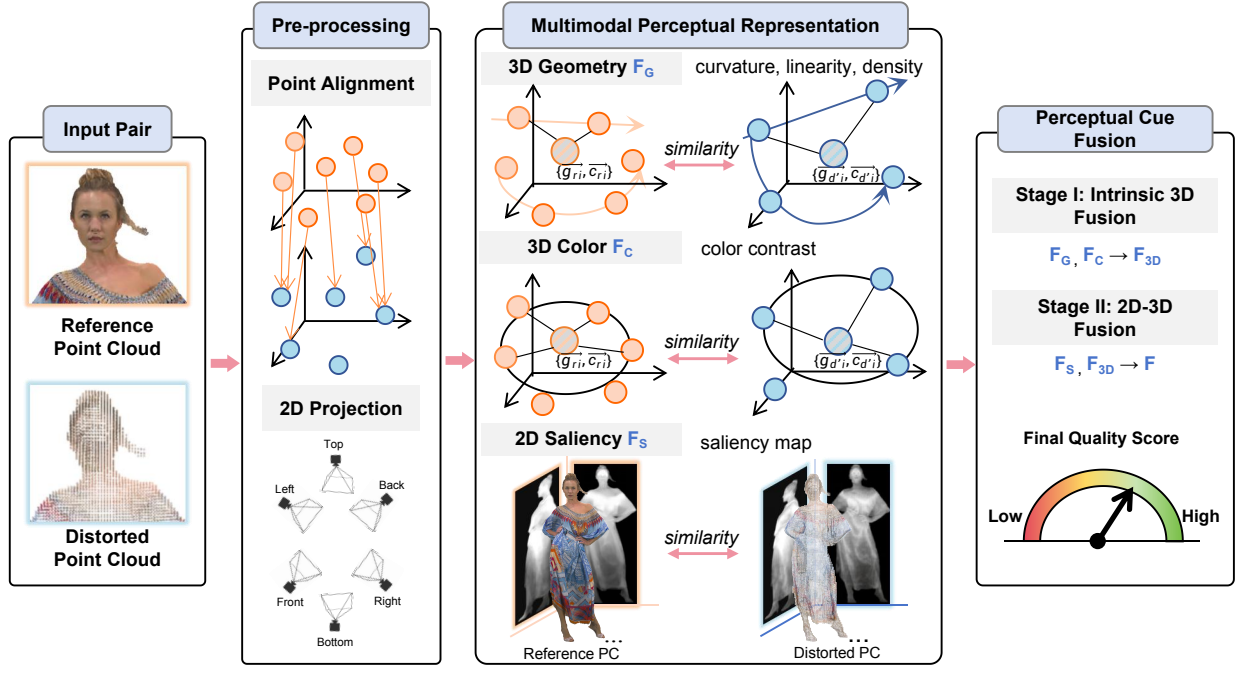


Fig. 1. Overview of the proposed MOSAIC framework from Marr's three-level perspective. The computational goal is to predict perceptual quality from discrepancies between reference and distorted point clouds. The representational level comprises geometric, color, and saliency cues. The implementational level realizes these cues through 3D feature extraction, 2D saliency estimation, and two-stage adaptive fusion.

$$\begin{aligned}
 G_{ri}^{(3)} &= \frac{1}{K_p} \sum_{k=1}^{K_p} \|\vec{g}_{ri}^k - \vec{g}_{ri}\|_2^2, \\
 G_{d'i}^{(3)} &= \frac{1}{K_p} \sum_{k=1}^{K_p} \|\vec{g}_{d'i}^k - \vec{g}_{d'i}\|_2^2.
 \end{aligned} \quad (3)$$

$\vec{g}_{ri}$ and $\vec{g}_{d'i}$ are the coordinates of the i -th point, and $\vec{g}_{ri}^k$ and $\vec{g}_{d'i}^k$ represent the coordinates of its k -th nearest neighbor; K_p is the number of nearest neighbors used to compute local density. Together with curvature and linearity, this density term makes the structural cue sensitive to both surface-shape changes and sampling-distribution changes, which are often affected differently by compression, downsampling, and noise.

2) *Appearance Representation Using Color Features*: MOSAIC uses the native RGB color space. Although alternative spaces (e.g., YUV or LAB) have been explored in prior studies [5], [10], RGB directly represents the stored color attributes in many point-cloud datasets and avoids additional color-space conversion. As reported in Table VIII, RGB also achieves the best overall performance within our framework. Since human perception is sensitive to local chromatic contrast, we extract a local RGB residual as the color feature for each point $\vec{p}_i$:

$$\mathbf{C}_{ri} = [C_{ri}^{(1)}, C_{ri}^{(2)}, C_{ri}^{(3)}] = \frac{1}{K_c} \sum_{k=1}^{K_c} (\vec{c}_{ri}^k - \vec{c}_{ri}), \quad (4)$$

$$\mathbf{C}_{d'i} = [C_{d'i}^{(1)}, C_{d'i}^{(2)}, C_{d'i}^{(3)}] = \frac{1}{K_c} \sum_{k=1}^{K_c} (\vec{c}_{d'i}^k - \vec{c}_{d'i}), \quad (5)$$

where $\mathbf{C}_{ri}$ and $\mathbf{C}_{d'i}$ denote the color features composed of R, G, B components for reference and the aligned distorted

point cloud, respectively; $\vec{c}_{ri}$ and $\vec{c}_{d'i}$ represent the color of the i -th point, while $\vec{c}_{ri}^k$ and $\vec{c}_{d'i}^k$ correspond to the colors of its k -th nearest neighbor. K_c denotes the number of neighboring points used to evaluate local color contrast.

3) *Attention-related Saliency Representation*: Distortions can alter visual saliency patterns, and artifacts in salient regions are more likely to affect perceived quality. Visual saliency is therefore a useful cue for point-cloud quality evaluation, as also explored in saliency-based PCQA work [36].

Following the orthogonal six-view projection (O6P) setting in [19], we first normalize both reference and distorted point clouds to a fixed projection grid and rasterize them onto the six faces of the axis-aligned bounding cube. These six perpendicular views cover the front/back, left/right, and top/bottom directions with deterministic canonical coverage. Saliency maps are then extracted from each projected view using the established image saliency detection algorithm [37], resulting in saliency representations S_r^v and S_d^v for reference and distorted point clouds, respectively, with $v \in \{1, \dots, 6\}$ as the view index.

We use projected 2D saliency for two reasons. First, human observers typically perceive rendered point clouds through image-like views, making view-based saliency a natural approximation of visual attention. Second, extracting saliency in the projected domain avoids the ambiguity of defining attention directly on irregular 3D point sets.

Unlike saliency-guided local metrics that transfer saliency back to 3D neighborhoods for local descriptor comparison and score pooling, MOSAIC keeps saliency in the projected-view domain and treats projected 2D saliency-map consistency as an independent modality-level fidelity cue. This choice is motivated by the observation that distortions may alter

the saliency distribution itself, which provides complementary information beyond geometry and color discrepancies.

C. Modality-wise Perceptual Fidelity Measurement

We then compute point-to-point similarity between the reference and aligned distorted point clouds for each geometry and color feature. The geometry and color fidelity scores F_G and F_C are obtained by averaging over N points and three features:

$$F_G = \frac{1}{3} \frac{1}{N} \sum_{l=1}^3 \sum_{i=1}^N \frac{2G_{ri}^{(l)} G_{d'i}^{(l)} + T}{(G_{ri}^{(l)})^2 + (G_{d'i}^{(l)})^2 + T}, \quad (6)$$

$$F_C = \frac{1}{3} \frac{1}{N} \sum_{l=1}^3 \sum_{i=1}^N \frac{2C_{ri}^{(l)} C_{d'i}^{(l)} + T}{(C_{ri}^{(l)})^2 + (C_{d'i}^{(l)})^2 + T}, \quad (7)$$

where T is a small constant used to avoid numerical instability.

For the saliency modality, we calculate the structural similarity index (SSIM) [38] for each view pair (S_r^v, S_d^v) , $v \in \{1, \dots, 6\}$. Following [36], the view scores are averaged to obtain the final saliency fidelity score F_S :

$$F_S = \frac{1}{6} \sum_{v=1}^6 \text{SSIM}(S_r^v, S_d^v). \quad (8)$$

D. Two-stage Perceptual Cue Fusion

After obtaining modality-wise fidelity scores, MOSAIC uses a two-stage fusion strategy that mirrors the layered organization of the proposed perceptual representation. Geometry and color describe intrinsic properties of the 3D object: geometry captures structural fidelity, while color captures appearance fidelity. These cues are first integrated into an intrinsic 3D fidelity representation. Saliency is derived from projected views and reflects the attentional importance of visible regions, so it is used in the second stage to modulate the intrinsic representation. This design reflects the intuition that perceptual quality depends not only on the amount of physical degradation, but also on whether visually important regions are affected.

$$F_{3D} = \frac{w_{F_G} F_G + w_{F_C} F_C}{w_{F_G} + w_{F_C} + \epsilon}, \quad (9)$$

$$F = \frac{w_{F_{3D}} F_{3D} + w_{F_S} F_S}{w_{F_{3D}} + w_{F_S} + \epsilon}, \quad (10)$$

where w_{F_G} , w_{F_C} , $w_{F_{3D}}$ and w_{F_S} denote the weights associated with geometry, color, fused 3D, and saliency fidelity scores, respectively; F_{3D} represents the fused intrinsic 3D fidelity score, F denotes the final perceptual fidelity score, and ϵ is a small constant.

Perceptual studies on multisensory cue integration show that the human visual system does not simply average all cues with fixed weights. Instead, the contribution of each cue is related to its reliability and uncertainty [39]. Human visual responses to stimulus intensity also tend to be nonlinear and saturated, and such behavior can be approximated by sigmoid-like functions [40]. These studies do not directly give a pooling formula for PCQA, but they suggest how the

weights should behave in our metric. A modality with higher fidelity should receive a larger weight because it is more consistent between the reference and distorted point clouds. The weight should also change smoothly with the fidelity score, so small score variations do not lead to abrupt quality changes. In addition, the weight should be bounded, so one highly consistent modality cannot completely dominate the fusion.

Based on this idea, MOSAIC treats each modality fidelity score, denoted as F_m below, as a simple cue-consistency measure. A low F_m is still kept in the fusion rather than removed. Since the final score is a weighted average of the fidelity values, this low-fidelity modality still lowers the final quality score. The sigmoid weight should therefore be read as cue reliability, not as distortion importance. A degraded cue is penalized through its lower fidelity value in the weighted average, while the sigmoid prevents an unstable low-consistency cue from taking over the fusion.

We therefore use a sigmoid function to map each fidelity score to a modality weight:

$$w_{F_m} = \frac{1}{1 + \exp(-\alpha F_m)}, \quad (11)$$

$$F_m \in \{F_G, F_C, F_{3D}, F_S\}.$$

where α controls the slope of the nonlinear mapping. We set $\alpha = 1$ in all experiments to keep the weighting stable and avoid extra dataset-specific tuning. Since the modality fidelity scores are normalized similarity measures, the sigmoid mapping is monotonic and maps each weight to $(0, 1)$. Thus, a more consistent cue receives a larger but bounded weight. The saturation of the sigmoid also reduces unnecessary weight changes when a cue already has high fidelity. Here, the sigmoid is not used as an exact model of neural responses. It is used as a simple regularized mapping for adaptive cue fusion. The ablation in Table VI and the parameter sensitivity analysis in Table XI further examine this choice.

IV. EXPERIMENTS

A. Experimental Settings

Datasets: We evaluate MOSAIC on five publicly accessible PCQA datasets, including SJTU [19], WPC [20], ICIP [41], LS-PCQA [42], and MJ-PCCD [43]. These datasets cover diverse distortion types and content characteristics, providing comprehensive benchmarks for evaluating prediction accuracy and generalization.

- The **SJTU dataset** [19] comprises 9 reference point clouds and 378 distorted versions generated using seven distortion types, including octree compression, color noise, geometry Gaussian noise, downsampling, and their combinations, applied at six distortion levels.
- The **WPC dataset** [20] consists of 20 reference point clouds, each subjected to five types of distortions, including Gaussian noise, downsampling, and compression noise generated by MPEG G-PCC (Trisoup), G-PCC (Octree), and V-PCC, producing 740 distorted samples.
- The **ICIP dataset** [41] contains six human-body point clouds compressed using MPEG G-PCC (Octree), G-PCC

TABLE II

PERFORMANCE EVALUATION OF TRAINING-FREE FR-PCQA METRICS ON DIFFERENT DATASETS. THE BEST, SECOND-BEST, AND THIRD-BEST RESULTS ARE MARKED IN **RED**, **GREEN**, AND **BLUE**, RESPECTIVELY. A DASH INDICATES THAT THE RESULT IS UNAVAILABLE OR NOT DIRECTLY REPORTED. AVERAGE VALUES ARE ARITHMETIC MEANS OVER DATASETS AND ARE REPORTED ONLY WHEN ALL DATASET RESULTS ARE AVAILABLE.

	SJTU [19]			WPC [20]			ICIP [41]			LS-PCQA [42]			MJ-PCCD [43]			Average		
	PLCC↑	SROCC↑	RMSE↓	PLCC↑	SROCC↑	RMSE↓	PLCC↑	SROCC↑	RMSE↓	PLCC↑	SROCC↑	RMSE↓	PLCC↑	SROCC↑	RMSE↓	PLCC↑	SROCC↑	RMSE↓
M-p2po [2]	0.728	0.801	1.717	0.578	0.566	18.714	0.947	0.935	0.365	0.398	0.185	0.763	0.434	0.314	1.054	0.617	0.560	4.523
M-p2pl [3]	0.652	0.661	1.893	0.454	0.446	20.422	0.954	0.948	0.341	0.351	0.166	0.779	0.396	0.370	1.074	0.561	0.518	4.902
PSNRyuv [4]	0.649	0.646	1.847	0.551	0.536	19.132	0.868	0.868	0.564	0.390	0.386	0.766	0.416	0.310	1.064	0.575	0.549	4.675
PCQM [5]	0.860	0.847	1.237	0.750	0.743	15.164	0.845	0.866	0.608	0.404	0.362	0.761	0.581	0.442	0.953	0.688	0.652	3.745
PointSSIM [23]	0.846	0.844	1.295	0.551	0.541	19.132	0.687	0.621	0.825	0.248	0.149	0.806	0.491	0.431	1.019	0.565	0.517	4.615
GraphSIM [6]	0.856	0.842	1.256	0.687	0.678	16.649	0.906	0.889	0.481	0.252	0.243	0.805	0.574	0.546	0.958	0.655	0.640	4.030
MS-GraphSIM [7]	0.899	0.890	1.031	0.709	0.704	16.172	0.482	0.021	1.001	0.327	0.307	0.786	—	—	—	—	—	—
MPED [24]	0.901	0.892	1.053	0.626	0.627	17.874	0.958	0.953	0.328	0.540	0.501	0.700	0.534	0.440	0.990	0.712	0.683	4.189
PQI [1]	0.912	0.906	0.996	0.813	0.808	13.364	0.966	0.955	0.292	0.437	0.379	0.748	0.582	0.481	0.952	0.742	0.706	3.270
TCDM [25]	0.930	0.910	0.895	0.806	0.804	13.557	0.942	0.935	0.383	0.335	0.325	0.784	0.885	0.837	0.546	0.780	0.762	3.233
PHM [26]	0.906	0.888	1.025	0.838	0.833	12.510	0.969	0.958	0.280	0.546	0.500	0.697	0.780	0.690	0.733	0.808	0.774	3.049
MOSAIC	0.912	0.898	0.994	0.844	0.834	12.292	0.965	0.955	0.297	0.622	0.598	0.652	0.831	0.755	0.651	0.835	0.808	2.977

TABLE III

PERFORMANCE EVALUATION ON DIFFERENT DISTORTION TYPES. THE BEST, SECOND-BEST, AND THIRD-BEST RESULTS ARE MARKED IN **RED**, **GREEN**, AND **BLUE**, RESPECTIVELY. AVERAGE VALUES ARE ARITHMETIC MEANS OVER THE FIVE DISTORTION GROUPS.

	G-PCC Octree			G-PCC Trisoup			V-PCC			Downsampling			Gaussian Noise			Average		
	PLCC↑	SROCC↑	RMSE↓	PLCC↑	SROCC↑	RMSE↓	PLCC↑	SROCC↑	RMSE↓	PLCC↑	SROCC↑	RMSE↓	PLCC↑	SROCC↑	RMSE↓	PLCC↑	SROCC↑	RMSE↓
M-p2po [2]	0.489	0.173	0.220	0.531	0.511	0.208	0.508	0.487	0.181	0.087	0.713	0.286	0.147	0.145	0.210	0.352	0.406	0.221
M-p2pl [3]	0.498	0.176	0.219	0.517	0.497	0.210	0.558	0.544	0.174	0.087	0.649	0.286	0.147	0.041	0.210	0.361	0.381	0.220
PSNRyuv [4]	0.700	0.675	0.180	0.618	0.621	0.193	0.464	0.382	0.186	0.631	0.631	0.223	0.729	0.682	0.145	0.628	0.598	0.185
PCQM [5]	0.886	0.838	0.117	0.791	0.792	0.150	0.718	0.698	0.146	0.886	0.840	0.133	0.893	0.869	0.096	0.835	0.807	0.128
PointSSIM [23]	0.449	0.134	0.225	0.567	0.562	0.203	0.606	0.592	0.167	0.714	0.701	0.201	0.747	0.710	0.141	0.617	0.540	0.187
GraphSIM [6]	0.830	0.781	0.141	0.785	0.785	0.152	0.499	0.436	0.182	0.951	0.905	0.089	0.876	0.853	0.102	0.788	0.752	0.133
MPED [24]	0.535	0.494	0.213	0.548	0.545	0.206	0.688	0.696	0.153	0.933	0.903	0.103	0.855	0.820	0.110	0.712	0.692	0.157
PQI [1]	0.776	0.743	0.159	0.737	0.742	0.166	0.582	0.560	0.171	0.948	0.913	0.091	0.916	0.901	0.085	0.792	0.772	0.134
TCDM [25]	0.686	0.619	0.183	0.846	0.845	0.121	0.744	0.742	0.141	0.943	0.894	0.096	0.897	0.884	0.094	0.823	0.797	0.127
PHM [26]	0.704	0.667	0.179	0.874	0.869	0.120	0.581	0.535	0.171	0.950	0.908	0.090	0.929	0.913	0.079	0.808	0.778	0.128
MOSAIC	0.798	0.766	0.152	0.819	0.816	0.141	0.781	0.769	0.131	0.948	0.909	0.092	0.906	0.872	0.090	0.850	0.826	0.121

(Trisoup), and V-PCC codecs at five bitrates, resulting in 90 impaired samples.

- The **LS-PCQA dataset** [42] includes 930 distorted point clouds created from 85 reference point clouds using 31 distortion types, where Part I is annotated with MOSs.
- The **MJ-PCCD dataset** [43] includes six reference point clouds and 213 distorted samples generated using G-PCC, V-PCC, and learning-based JPEG Pleno standard codecs.

Implementation Details: For neighborhood construction, $K_g = 100$, $K_p = 10$, and $K_c = 10$ are used for geometry (Eq. (1)), point density (Eq. (3)), and color feature (Eqs. (4) and (5)) extraction, respectively. MOSAIC adopts KNN-based neighborhoods rather than radius-based neighborhoods (see Sec. IV-D2) to balance structural coverage and computational cost. Following prior studies [34], [35], a relatively large K_g is used for local shape estimation, while smaller neighborhoods are used for density and color to preserve fine-scale point-spacing changes and local chromatic contrast. The nonlinear fusion parameter α is set to 1 in Eq. (11).

Evaluation Criteria: For all methods, predicted scores are mapped to MOS values using the five-parameter logistic function recommended by the ITU VQEG (Video Quality Expert Group) [44]:

$$\hat{y} = \beta_1 \left(0.5 - \frac{1}{1 + e^{\beta_2(y - \beta_3)}} \right) + \beta_4 y + \beta_5, \quad (12)$$

where y represents the predicted quality score, and $\beta_1, \dots, \beta_5$ are optimized by minimizing the squared errors between the

mapped scores $\hat{y}$ and the subjective MOS values. We report Pearson Linear Correlation Coefficient (PLCC), Spearman Rank Correlation Coefficient (SROCC), and root mean square error (RMSE), which measure prediction accuracy, monotonicity, and error, respectively. Higher PLCC and SROCC and lower RMSE indicate better performance. For the Average columns, PLCC, SROCC, and RMSE are averaged only when all corresponding dataset results are available. These averages provide an overview of whether a metric remains stable across several databases. For RMSE, the average should still be read together with dataset-wise RMSE because its absolute value depends on the subjective-score scale and logistic fitting range of each database.

B. Comparisons with Training-free Metrics

We compare MOSAIC with MSE-p2point (M-p2po) [2], MSE-p2plane (M-p2pl) [3], PSNRyuv [4], PointSSIM [23], PCQM [5], GraphSIM [6], MS-GraphSIM [7], MPED [24], PQI [1], TCDM [25], and PHM [26]. For training-free metrics, objective scores are evaluated directly on each dataset.

1) *Dataset-wise Evaluation:* Table II shows that no single metric dominates all datasets. TCDM gives the best results on SJTU and MJ-PCCD, while PHM is particularly strong on ICIP. The Average columns provide a compact summary over datasets with complete results and show that MOSAIC achieves the best average PLCC/SROCC/RMSE. MOSAIC also obtains the best performance on WPC and LS-PCQA, with (PLCC/SROCC/RMSE) values of (0.844/0.834/12.292)

TABLE IV
RUNTIME COMPARISON ACROSS DIFFERENT METHODS.

Method	PSNR _{Yuv} [4]	PCQM [5]	PointSSIM [23]	GraphSIM [6]	MPED [24]	PQI [1]	TCDM [25]	PHM [26]	MOSAIC
Time (s)	10.30	3.55	4.24	11.68	7.60	4.71	5.22	14.96	6.41

and (0.622/0.598/0.652), respectively. These two datasets are especially informative for generalization, as they are fairly large scale and challenging [20], [42].

For the datasets where MOSAIC is not the best method, its performance remains close to the leading result. On MJ-PCCD, MOSAIC ranks second on all three criteria and clearly improves over PHM and PQI. On SJTU, it ties PQI in PLCC and gives the second-best RMSE. On ICIP, PHM and PQI are slightly better, which is consistent with their strong behavior on compression-oriented databases.

2) *Distortion-wise Evaluation*: To evaluate robustness under different distortion mechanisms, we conduct a distortion-wise analysis by integrating samples from SJTU [19], WPC [20], ICIP [41], and MJ-PCCD [43]. Five representative distortion types are considered. G-PCC Octree Compression includes 182 distorted point clouds, with 80 samples from WPC, 30 from ICIP, and 72 from MJ-PCCD. G-PCC Trisoup Compression includes 270 samples, with 240 from WPC and 30 from ICIP. V-PCC Compression includes 279 samples, with 180 from WPC, 30 from ICIP, and 69 from MJ-PCCD. Downsampling includes 114 samples, with 54 from SJTU and 60 from WPC. Gaussian Noise includes 234 samples, with 54 from SJTU and 180 from WPC.

Table III shows that MOSAIC remains in the leading group across different distortion mechanisms. The Average columns show the best average PLCC/SROCC/RMSE across the five distortion groups, while individual categories reveal complementary strengths among PCQM, TCDM, PHM, PQI, and MOSAIC. This behavior indicates good cross-distortion generalization: MOSAIC does not rely on distortion-specific switching or supervised fitting, but still remains competitive under compression, downsampling, and noise distortions.

The main exception is G-PCC Octree, where PCQM performs best. The gap mainly appears in lossless-geometry and lossy-attribute cases, which is consistent with the metrics' formula designs. Following its original formulation, PCQM represents local geometry/color evidence as distance-based distortion features [5]. The final score is a linear combination, $PCQM = \sum_i w_i f_i$; when geometry is nearly unchanged, it is driven mainly by attribute-distance terms. By contrast, TCDM [25] builds geometry- and color-channel complexity similarities from self- and transformational-complexity terms, multiplies them, and then combines this product with a prediction-based feature. PHM [26] multiplicatively fuses visible-difference and appearance-degradation terms, and MOSAIC averages normalized geometry, color, and saliency fidelities with bounded adaptive weights. These similarity- or product-style formulations are generally robust, but they may be less discriminative for subtle attribute-only changes when the geometry cue remains almost perfect. This distance-versus-similarity behavior deserves further study.

TABLE V
PLCC/SROCC COMPARISON BETWEEN MOSAIC AND REPRESENTATIVE REGRESSION- OR LEARNING-BASED FR-PCQA REFERENCES ON OVERLAPPING DATASETS.

Method	SJTU [19]	WPC [20]	ICIP [41]	MJ-PCCD [43]
MOSAIC	0.912/0.898	0.844/0.834	0.965/0.955	0.831/0.755
PointPCA [9]	0.909/0.879	0.891/0.885	–	–
FRSVR [10]	–	0.811/0.796	0.956/0.935	0.709/0.620
SGW-PCQA+ [11]	–	0.873/0.869	–	–
FR-MLLM [13]	0.935/0.915	0.889/0.881	–	–
CT-PCQA (FR) [12]	0.929/0.910	0.881/0.878	–	–

These results indicate that MOSAIC, TCDM, and PHM form a leading group of recent training-free FR-PCQA metrics. A common reason is that all three revisit PCQA from a new perceptual or computational perspective, rather than only refining conventional point-wise errors. TCDM views quality degradation through transformational complexity. PHM uses a hybrid strategy to model different HVS responses at different distortion levels. MOSAIC treats FR-PCQA as multimodal cue fusion among 3D structure, 3D appearance, and projected 2D saliency. Within this group, the value of MOSAIC lies in balanced generalization across datasets and distortion types through an interpretable two-stage fusion design, without supervised fitting or distortion-specific switching.

3) *Computational Complexity*: We evaluate computational efficiency on the SJTU dataset. Because the compared methods are implemented in different software environments (C/C++, MATLAB, and Python), the reported runtime is intended as a practical reference rather than a strict speed benchmark. As shown in Table IV, PCQM, PointSSIM, PQI, and TCDM are relatively efficient, whereas PHM is slower because of graph construction, graph-domain analysis, and spectral operations. MOSAIC requires 6.41 seconds per distorted point cloud on average, which is lower than PSNR_{Yuv}, GraphSIM, MPED, and PHM.

C. Comparison with Regression- & Learning-Based Methods

Regression-based and learning-based FR-PCQA methods are useful references, but we keep them separate from the main training-free comparison in Table II. Their reported scores may also reflect subjective-label fitting, evaluation splits, learned representations, or SVR-based score integration, so Table V is used to provide context rather than a strict ranking.

Table V uses the best PLCC/SROCC values reported in the original papers and keeps only datasets shared with our evaluation. The comparison is therefore restricted to shared datasets, although the reported values still come from each paper's own protocol. Under this practical comparison, MOSAIC is competitive with regression- and learning-based references. It

slightly outperforms PointPCA on SJTU and FRSVR on WPC, ICIP, and MJ-PCCD, while remaining close to SGW-PCQA+, FR-MLLM, and CT-PCQA on SJTU/WPC. These methods can benefit from subjective-label fitting or learned representations, whereas MOSAIC uses deterministic training-free cues.

D. Ablation Studies

This section probes the potential of MOSAIC by isolating its internal design choices, including multimodal cue fusion, neighborhood construction, color space, projection strategy, keypoint selection, and parameter sensitivity.

TABLE VI
MULTIMODAL CUE FUSION STRATEGIES.

Scheme	Fusion	WPC [20]			LS-PCQA [42]		
		PLCC↑	SROCC↑	RMSE↓	PLCC↑	SROCC↑	RMSE↓
Only 3D	Adaptive	0.793	0.775	13.973	0.523	0.493	0.709
One-stage	Uniform	0.835	0.823	12.622	0.564	0.533	0.687
	Adaptive	0.836	0.825	12.573	0.586	0.560	0.674
Two-stage	Uniform	0.844	0.833	12.296	0.592	0.566	0.671
	Adaptive	0.844	0.834	12.292	0.622	0.598	0.652

1) *Multimodal Cue Fusion*: Table VI evaluates the strategy on how to fuse multimodal cues. The “Only 3D” baseline uses intrinsic 3D geometry and color cues without projected 2D saliency. The one-stage variants combine all cues in a single step, whereas the two-stage variants first fuse geometry and color into an intrinsic 3D fidelity score and then incorporate saliency. Uniform and adaptive denote linear averaging and sigmoid-based weighting, respectively. We choose WPC and LS-PCQA for this ablation because they are large-scale and challenging, and they provide a clear view of the contribution of each design choice.

Replacing the “Only 3D” variant with multimodal fusion improves PLCC/SROCC from 0.793/0.775 to 0.844/0.834 on WPC, and from 0.523/0.493 to 0.622/0.598 on LS-PCQA. This indicates that projected 2D saliency provides perceptual information that is not fully captured by intrinsic 3D geometry and color fidelity.

The fusion organization also matters. Two-stage variants consistently outperform one-stage variants, and the best result is obtained by two-stage adaptive fusion. The gain is especially clear on LS-PCQA, where adaptive weighting in the two-stage form improves PLCC/SROCC/RMSE from 0.592/0.566/0.671 to 0.622/0.598/0.652. This supports the proposed cue organization: intrinsic 3D geometric structure and color appearance first form a fidelity representation, which is then modulated by 2D saliency. It therefore suggests that MOSAIC’s potential comes from the order and interaction of perceptual cues, not simply from adding more features.

2) *Neighborhood Construction*: In the local shape feature extraction (Eq. (1)), neighborhoods can be constructed using either KNN-based or radius-based strategies. Table VII shows that KNN-based neighborhoods generally provide more stable performance across datasets, while radius-based neighborhoods are comparable only in some cases. Runtime is also important for large-scale point clouds. In our implementation, radius-based neighborhood construction incurs substantially

TABLE VII
ON NEIGHBORHOOD CONSTRUCTION FOR LOCAL SHAPE FEATURES. EACH CELL REPORTS PLCC/SROCC/RMSE.

Dataset	KNN	Radius
SJTU [19]	0.912/0.898/0.994	0.880/0.873/1.154
WPC [20]	0.844/0.834/12.292	0.826/0.817/12.911
ICIP [41]	0.965/0.955/0.297	0.964/0.954/0.302
LS-PCQA [42]	0.622/0.598/0.652	0.589/0.566/0.672
MJ-PCCD [43]	0.831/0.755/0.651	0.838/0.753/0.638

higher search cost, with an average runtime approximately $16\times$ higher than the KNN-based strategy on SJTU dataset. MOSAIC therefore adopts KNN-based neighborhoods to balance prediction accuracy and computational efficiency.

3) *Color Space*: MOSAIC adopts RGB because it is consistent with the stored color attributes of point-cloud datasets. We also evaluate MOSAIC using YUV and CIE-LAB (LAB), as reported in Table VIII. RGB achieves the best overall performance on most datasets. Although YUV and LAB are slightly better on LS-PCQA, these gains are marginal and come with noticeable degradation on other datasets. Moreover, using RGB avoids the need for color space conversion, which can introduce additional computational cost and potential numerical errors. We therefore keep RGB as the default color representation in this work.

4) *Projection Protocol*: Table IX examines how the projected 2D saliency cue is affected when only the 3D-to-2D view-generation protocol is changed. In all variants, MOSAIC uses the RGB projection images for saliency extraction and keeps the subsequent saliency comparison and cue fusion unchanged. The three protocols are selected as representative view-generation families in projection-based PCQA: O6P represents deterministic fixed-grid orthogonal projection, FVC represents static virtual-camera rendering, and MCV represents trajectory-based moving-camera rendering.

- **O6P** (orthogonal six-view projection) [19] normalizes the point cloud to a fixed projection grid and rasterizes it onto the six faces of the axis-aligned bounding cube, namely front/back, left/right, and top/bottom. This coordinate-grid projection gives deterministic coverage without virtual-camera placement. O6P is the default projection protocol in MOSAIC.
- **FVC** (fixed virtual-camera projection) [45] renders RGB views from fixed virtual cameras around the point cloud. The point cloud is normalized before rendering, and the virtual cameras keep a fixed viewing distance so that the

TABLE VIII
PERFORMANCE WITH VARIOUS COLOR SPACES. EACH CELL REPORTS PLCC/SROCC/RMSE. THE BEST VALUES WITHIN EACH DATASET AND METRIC ARE MARKED IN **BOLD**.

Dataset	RGB	YUV	LAB
SJTU [19]	0.912/0.898/0.994	0.882/0.849/1.142	0.856/0.817/1.256
WPC [20]	0.844/0.834/12.292	0.799/0.786/13.775	0.756/0.741/15.001
ICIP [41]	0.965/0.955/0.297	0.940/0.934/0.387	0.882/0.890/0.534
LS-PCQA [42]	0.622/0.598/0.652	0.651/0.638/0.632	0.640/ 0.645 /0.640
MJ-PCCD [43]	0.831/0.755/0.651	0.781/0.734/0.731	0.741/0.676/0.785

TABLE IX

PROJECTION PROTOCOLS FOR THE SALIENCY CUE. AVERAGE VALUES ARE ARITHMETIC MEANS OVER THE FIVE DATASETS; BEST VALUES ARE IN **BOLD**.

	SJTU [19]			WPC [20]			ICIP [41]			LS-PCQA [42]			MJ-PCCD [43]			Average		
	PLCC $\uparrow$	SROCC $\uparrow$	RMSE $\downarrow$	PLCC $\uparrow$	SROCC $\uparrow$	RMSE $\downarrow$	PLCC $\uparrow$	SROCC $\uparrow$	RMSE $\downarrow$	PLCC $\uparrow$	SROCC $\uparrow$	RMSE $\downarrow$	PLCC $\uparrow$	SROCC $\uparrow$	RMSE $\downarrow$	PLCC $\uparrow$	SROCC $\uparrow$	RMSE $\downarrow$
O6P [19]	0.912	0.898	0.994	0.844	0.834	12.292	0.965	0.955	0.297	0.622	0.598	0.652	0.831	0.755	0.651	0.835	0.808	2.977
FVC [45]	0.899	0.885	1.063	0.817	0.804	13.204	0.969	0.957	0.281	0.632	0.607	0.644	0.740	0.656	0.787	0.811	0.782	3.196
MCV [46]	0.894	0.881	1.086	0.810	0.798	13.431	0.966	0.953	0.293	0.636	0.608	0.642	0.696	0.629	0.840	0.800	0.774	3.258

TABLE X

ON KEYPOINT SELECTION. AVERAGE VALUES ARE ARITHMETIC MEANS OVER THE FIVE DATASETS; BEST VALUES ARE IN **BOLD**.

		SJTU [19]			WPC [20]			ICIP [41]			LS-PCQA [42]			MJ-PCCD [43]			Average		
Method	Keypoints	PLCC $\uparrow$	SROCC $\uparrow$	RMSE $\downarrow$	PLCC $\uparrow$	SROCC $\uparrow$	RMSE $\downarrow$	PLCC $\uparrow$	SROCC $\uparrow$	RMSE $\downarrow$	PLCC $\uparrow$	SROCC $\uparrow$	RMSE $\downarrow$	PLCC $\uparrow$	SROCC $\uparrow$	RMSE $\downarrow$	PLCC $\uparrow$	SROCC $\uparrow$	RMSE $\downarrow$
All	N	0.912	0.898	0.994	0.844	0.834	12.292	0.965	0.955	0.297	0.622	0.598	0.652	0.831	0.755	0.651	0.835	0.808	2.977
FPS [26]	$N/1000$	0.745	0.744	1.620	0.751	0.745	15.149	0.816	0.822	0.656	0.647	0.633	0.635	0.828	0.670	0.656	0.757	0.723	3.743
RS	400	0.742	0.739	1.628	0.752	0.748	15.105	0.818	0.823	0.653	0.641	0.623	0.638	0.824	0.661	0.663	0.755	0.719	3.737
FastFPS	400	0.732	0.732	1.653	0.753	0.749	15.088	0.821	0.828	0.650	0.641	0.622	0.639	0.819	0.654	0.672	0.753	0.717	3.740
HGS [6]	$N/10000$	0.716	0.719	1.695	0.740	0.735	15.413	0.806	0.814	0.673	0.634	0.611	0.643	0.782	0.613	0.730	0.736	0.698	3.831

TABLE XI

ONE-AT-A-TIME PARAMETER SENSITIVITY ANALYSIS. AVERAGE VALUES ARE ARITHMETIC MEANS OVER THE FIVE DATASETS.

		SJTU [19]			WPC [20]			ICIP [41]			LS-PCQA [42]			MJ-PCCD [43]			Average		
		PLCC $\uparrow$	SROCC $\uparrow$	RMSE $\downarrow$	PLCC $\uparrow$	SROCC $\uparrow$	RMSE $\downarrow$	PLCC $\uparrow$	SROCC $\uparrow$	RMSE $\downarrow$	PLCC $\uparrow$	SROCC $\uparrow$	RMSE $\downarrow$	PLCC $\uparrow$	SROCC $\uparrow$	RMSE $\downarrow$	PLCC $\uparrow$	SROCC $\uparrow$	RMSE $\downarrow$
Default		0.912	0.898	0.994	0.844	0.834	12.292	0.965	0.955	0.297	0.622	0.598	0.652	0.831	0.755	0.651	0.835	0.808	2.977
$K_g = 10$		0.892	0.882	1.096	0.846	0.837	12.231	0.969	0.964	0.279	0.628	0.604	0.648	0.807	0.729	0.691	0.828	0.803	2.989
$K_g = 50$		0.912	0.896	0.997	0.842	0.832	12.364	0.969	0.962	0.281	0.624	0.601	0.650	0.825	0.745	0.661	0.834	0.807	2.991
$K_p = 50$		0.902	0.890	1.047	0.832	0.822	12.726	0.968	0.958	0.285	0.614	0.586	0.657	0.865	0.774	0.587	0.836	0.806	3.060
$K_p = 100$		0.901	0.889	1.053	0.831	0.822	12.752	0.968	0.957	0.285	0.609	0.582	0.660	0.871	0.775	0.576	0.836	0.805	3.065
$K_c = 50$		0.897	0.876	1.074	0.842	0.835	12.376	0.960	0.947	0.319	0.614	0.600	0.657	0.815	0.754	0.678	0.826	0.802	3.021
$K_c = 100$		0.876	0.849	1.170	0.834	0.828	12.642	0.952	0.942	0.349	0.606	0.596	0.661	0.801	0.744	0.700	0.814	0.792	3.104
$\alpha = 0.5$		0.912	0.898	0.994	0.844	0.834	12.285	0.965	0.955	0.297	0.607	0.586	0.661	0.830	0.754	0.652	0.832	0.805	2.978
$\alpha = 3$		0.912	0.898	0.994	0.844	0.834	12.303	0.965	0.955	0.297	0.640	0.612	0.639	0.831	0.754	0.651	0.838	0.811	2.977

projected texture scale remains stable.

- **MCV** (moving-camera video projection) [46] generates frames by rotating the camera around the point cloud along circular trajectories, forming video clips for spatial-temporal VQA feature extraction; here, only the sampled RGB frames are used as saliency views.

O6P gives the most balanced performance: it is best on SJTU, WPC, and MJ-PCCD and also obtains the best Average PLCC/SROCC/RMSE of 0.835/0.808/2.977. FVC slightly improves ICIP, and MCV slightly improves LS-PCQA, indicating that alternative view-generation rules may fit certain content distributions. However, these gains are not stable across datasets when the rest of MOSAIC is kept unchanged. We therefore keep O6P as the default projection protocol because it gives the strongest average performance and uses deterministic fixed-grid coverage.

5) *Keypoint Selection*: Table X compares five sampling protocols while keeping all other parts of MOSAIC unchanged. **All** is the default setting in MOSAIC and uses every reference point. **FPS** uses farthest point sampling with the $N/1000$ seed ratio adopted in PHM's Voronoi-patch construction [26]. **RS** randomly selects 400 seeds, and **FastFPS** first forms a random candidate subset and then applies FPS; these two rows are 400-seed control variants motivated by the seed budget used in TCDM's Voronoi partition [25]. **HGS** uses the high-pass graph sampling idea from GraphSIM [6] to select geometry-aware keypoints, here with $N/10000$ sampled points.

Using all points gives the best performance on SJTU, WPC, ICIP, and MJ-PCCD, and the Average columns further

show its overall advantage. Sparse keypoint strategies slightly improve LS-PCQA, but they substantially degrade the other four datasets. This behavior is consistent with the role of sampling in the original methods: graph or Voronoi metrics can use sparse seeds as patch centers, whereas MOSAIC measures point-aligned local geometry, color, and saliency fidelities. If too many points are removed, local color variation, density changes, and attention-related evidence may be lost. The slight LS-PCQA gain suggests that sparse seeds can sometimes suppress noisy fine-scale matches, but the overall results favor full point coverage. Since the all-point setting still presents moderate runtime in Table IV, it is used as the default strategy in this work.

6) *Parameter Sensitivity*: To examine whether MOSAIC relies on parameter tuning, we conduct a one-at-a-time sensitivity analysis for K_g , K_p , K_c , and α , as reported in Table XI. In each non-default row, only the parameter named in the first column is changed, while the other three parameters remain fixed at the default setting. This design is intentional. The goal is not to search for the best parameter combination, but to test whether the proposed cue extraction and fusion remain stable when one design factor is perturbed. A joint search over K_g , K_p , K_c , and α would mix the effects of local shape scale, density scale, color scale, and fusion slope, and could introduce dataset-specific tuning that is inconsistent with the training-free setting.

The default setting remains competitive across datasets, indicating that the metric is not highly sensitive to moderate parameter changes. Reducing the geometry neighborhood

size K_g to 10 or 50 causes only limited fluctuations on most datasets, suggesting that the structural representation is stable once sufficient local geometry is captured. Increasing the density neighborhood size K_p improves MJ-PCCD but degrades SJTU, WPC, and LS-PCQA, indicating that overly large density neighborhoods may smooth fine-scale sampling changes. A similar effect appears for the color neighborhood size K_c , where larger neighborhoods dilute local chromatic differences. For the nonlinear mapping parameter, $\alpha = 3$ gives slightly higher average PLCC/SROCC, mainly due to the LS-PCQA gain, while $\alpha = 1$ remains nearly tied in average RMSE and avoids a sharper weighting curve. We therefore use the default parameters for all datasets.

V. CONCLUSION

This paper investigated FR-PCQA from the perspective of multimodal perceptual cue organization. The main insight is that 3D geometry and color describe how the point cloud itself is degraded, whereas projected 2D saliency indicates whether these degradations are likely to be noticed in rendered views. In MOSAIC, local geometric structure and color appearance are therefore first integrated into an intrinsic 3D fidelity representation, and projected 2D saliency consistency is then incorporated as a complementary view-dependent cue. Experiments on five public PCQA datasets show that this training-free formulation is sufficient to place MOSAIC in the leading group of recent FR-PCQA metrics, with strong average performance, clear gains on WPC and LS-PCQA, and competitive results against representative learning-based references without MOS-supervised score fitting.

The current formulation also has several scope limitations. First, the saliency branch uses fixed six-view orthogonal projections, which keeps the metric deterministic but does not model user-specific viewing trajectories, dynamic head motion, or viewpoint-dependent occlusion changes in interactive rendering. Second, the neighborhood sizes and fusion slope are fixed across all datasets to preserve reproducibility, although point clouds with very different densities, scales, or acquisition pipelines may benefit from adaptive parameter selection. Future work will therefore explore view-adaptive or trajectory-aware saliency cues, density-aware neighborhood construction, and more efficient implementations for large-scale point clouds.

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Jiaxin Li Jiaxin Li is with the School of Electronic Science and Engineering, Nanjing University, Nanjing, China. Jiaxin Li's research interests include point cloud processing, visual quality assessment, and immersive visual media.

Tong Chen Tong Chen is with the School of Electronic Science and Engineering, Nanjing University, Nanjing, China. Tong Chen's research interests include multimedia signal processing, point cloud quality assessment, and perceptual visual computing.

Zehong Li Zehong Li is with Hangzhou Normal University, Hangzhou, China. Zehong Li's research interests include point cloud analysis, visual quality modeling, and 3D media processing.

Dandan Ding Dandan Ding is with Hangzhou Normal University, Hangzhou, China. Dandan Ding's research interests include visual perception, multimedia quality assessment, and immersive media processing.

Zhan Ma Zhan Ma is with the School of Electronic Science and Engineering, Nanjing University, Nanjing, China. Zhan Ma's research interests include image and video processing, point cloud coding, visual quality assessment, and multimedia communication.